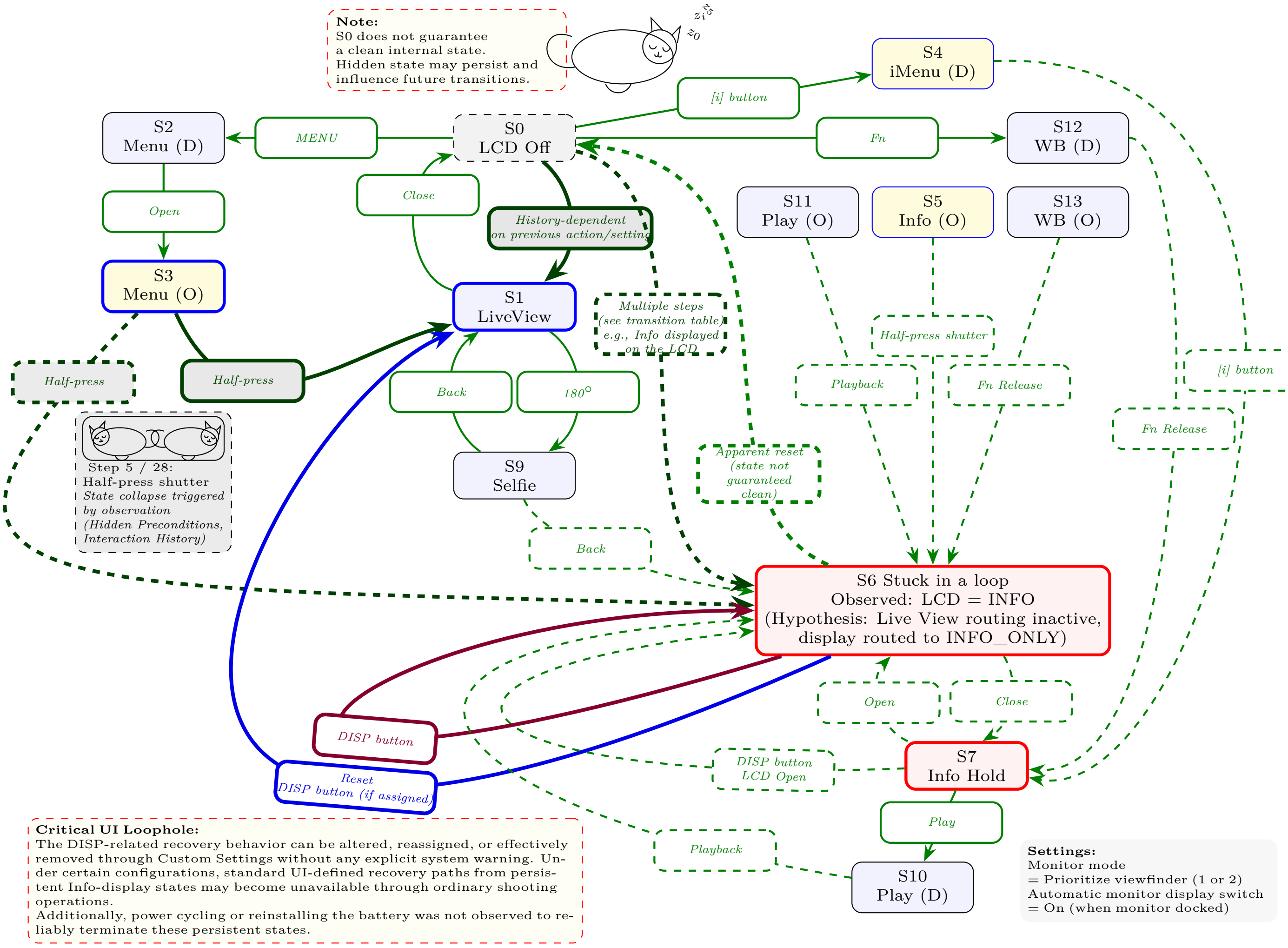
Deterministic Failure in Display State Transition of Z f (C: Ver. 3.01)

"CASE-A1: Schrödinger's Sisyphean Loop" *



Note: The state labels and step numbers used in this diagram correspond to the definitions introduced in "CASE-A1: Schrödinger's Sisyphean Loop (Rev. 1.6)." They are reused here for conceptual continuity and do not represent an exhaustive or formally complete state model.

Technical Observations

- This diagram is based on the observed behavior of a specific Nikon Z f unit (firmware v3.01) and represents a non-exhaustive set of user-observable state transitions. Consistent behavior across all units or environments is not guaranteed. Actual transitions may depend on previous interaction history and cannot be fully determined by visible states alone.

- Nikon has repeatedly described the relevant Info-display behavior as part of the product specification. In addition, through later support correspondence, Nikon support confirmed several individual real-camera behaviors related to Display 5 / Info persistence, including cases in which shutter-button half-press does not restore Live View, cases in which the Info display persists across power cycling, and cases in which disabling the DISP recovery path can leave the camera without an ordinary immediate recovery operation.

- **Reproduction parameters:** For the specific menu settings required to reproduce each transition (e.g., Step 9, 28, 37, 45), please refer to **Case A1: Schrödinger's Sisyphean Loop**. For later Nikon support confirmations, please refer to the support confirmation documents in the repository.

- **Indeterminacy of state transitions:** The transition from Step 3 is not uniquely determined by the external input alone; it is dependent on internal state history that is not externally observable. Under these conditions, even if two cameras appear to have identical settings and display states, their behavior may diverge due to differences in internal state. As a result, when encountering a shooting opportunity that requires Live View, it is not possible to determine in advance which camera will reliably allow capture.

  For example, in Step 5 the system transitions to S1 (Live View), whereas in Step 28 it transitions into the S6 (Info display) loop. Thus, the outcome is not uniquely determined for the same input, meaning that whether shooting is possible cannot be known prior to operation.

- **Loop behavior and priority:** This camera provides several monitor modes, including "Automatic display switching," "Viewfinder only," "Monitor only," and "Prioritize viewfinder (1, 2)." However, once the system enters the dashed loop region centered on S6, it cannot return to Live View through the standard user input path defined within the normal state transition graph. In this state, even if the user has selected "Prioritize viewfinder," the behavior effectively becomes equivalent to "Viewfinder only," and Live View on the monitor is no longer available.

- **Recovery paths and practical lock behavior:** Recovery from the persistent Info-display state was observed through DISP-related display cycling while the LCD monitor remained active (e.g., S6 with the LCD open). When the LCD was closed (S7), the same input did not produce a recovery transition.

  Additional recovery paths were later observed, including: (i) reassigning the "DISP Cycle view info display" function to a custom button and then pressing it, and (ii) disabling "Display 5" in the custom monitor shooting display settings.

  However, these are mitigation paths rather than a complete resolution of the underlying display-state behavior. In particular, under configurations where no DISP-related recovery path remains accessible through ordinary shooting operations, Nikon support confirmed an extended sequence in which the camera remained on the Info display through repeated half-press operations, MENU return, and power cycling.

  Power cycling or removing and reinserting the battery was not observed to reliably terminate this state. **Reference Video: zf info lock (YouTube)**

---

*Etymology: Named after Schrödinger's cat to describe the non-deterministic state collapse triggered by shutter "observation," and the myth of Sisyphus to reflect the futile loop of Info displays that prevents a return to Live View.